

KAPFENBERG, Austria

WMO: 111760

Lat: 47.458N

Lon: 15.336E

Elev: 1696

StdP: 13.82

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.0	14.2	5.9	7.9	12.4	9.6	9.6	17.8	11.4	39.1	9.4	36.4	1.9	80	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	87.3	68.0	84.1	67.0	80.7	65.6	69.8	83.2	68.2	80.7	66.8	77.8	4.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	100.7	73.9	64.1	95.4	72.3	62.9	91.3	71.0	34.8	83.4	33.4	80.7	32.3	77.9	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.4	7.6	6.6	DB	5.7	92.8	6.0	2.3	1.4	94.5	-2.1	95.8	-5.5	97.1	-9.8	98.8
			WB	5.2	72.9	5.7	0.9	1.0	73.5	-2.3	74.0	-5.5	74.5	-9.7	75.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	29.3	34.1	41.6	49.9	57.1	64.7	67.5	66.5	58.3	50.1	40.3	31.0	(6)
	DBStd	14.78	7.22	7.04	6.92	6.33	6.20	6.33	5.69	5.49	6.10	6.57	6.70	6.73	(7)
	HDD50	2420	643	445	267	78	11	1	0	0	6	82	297	590	(8)
	HDD65	6046	1108	865	726	454	254	80	40	49	213	462	740	1055	(9)
	CDD50	2158	0	0	6	74	232	443	544	511	256	85	7	0	(10)
	CDD65	307	0	0	0	0	10	72	118	94	13	0	0	0	(11)
	CDH74	2848	0	0	0	18	156	708	1049	821	94	2	0	0	(12)
	CDH80	832	0	0	0	0	29	204	333	256	10	0	0	0	(13)
(14) Wind	WSAvg	2.7	2.5	2.7	3.0	3.0	3.0	2.9	2.8	2.6	2.4	2.3	2.3	2.4	(14)
(15) Precipitation	PrecAvg	36.7	1.4	1.5	2.1	2.3	3.9	5.0	5.1	5.0	3.6	2.8	2.3	1.8	(15)
	PrecMax	46.6	3.0	3.9	6.0	4.8	8.3	7.6	10.9	8.5	5.9	6.4	4.6	3.9	(16)
	PrecMin	25.0	0.1	0.1	0.3	0.4	1.7	2.4	1.5	2.2	1.2	0.4	0.0	0.3	(17)
	PrecStd	4.7	0.8	1.0	1.1	1.0	1.4	1.4	2.1	1.8	1.1	1.5	1.0	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	58.2	67.5	76.8	84.5	89.6	91.2	91.2	81.1	72.0	62.2	50.4	(19)
		MCWB	41.6	45.9	51.2	57.6	64.4	68.4	68.9	69.0	65.3	60.4	53.0	44.4	(20)
	2%	DB	44.0	51.6	63.0	71.1	78.3	85.6	87.0	86.7	76.8	66.8	56.0	45.3	(21)
		MCWB	39.6	43.1	49.9	54.7	62.1	67.4	68.4	68.1	63.4	57.7	51.2	40.1	(22)
	5%	DB	40.9	47.2	58.7	67.4	74.4	81.9	83.7	82.3	72.8	62.7	52.4	41.5	(23)
		MCWB	37.4	40.3	47.7	53.1	59.9	65.7	67.3	66.8	61.4	55.6	48.5	38.6	(24)
	10%	DB	38.1	43.5	54.2	63.2	70.1	78.0	80.3	78.5	69.2	59.2	49.3	39.1	(25)
		MCWB	35.2	38.6	45.4	51.3	57.8	63.8	65.8	65.3	59.5	53.8	46.3	36.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	47.6	52.7	58.6	65.7	71.3	71.8	71.2	66.4	61.7	54.4	45.9	(27)
		MCDB	46.8	56.2	64.7	74.8	79.4	84.1	84.7	85.5	77.6	70.4	59.1	50.0	(28)
	2%	WB	40.4	43.9	50.5	55.7	63.5	68.8	70.0	69.1	64.4	58.8	51.8	41.4	(29)
		MCDB	43.3	50.1	61.4	68.4	75.5	81.6	82.8	83.0	74.0	64.9	55.8	43.4	(30)
	5%	WB	37.9	41.2	48.4	54.0	61.4	66.9	68.5	67.5	62.8	56.5	49.1	39.0	(31)
		MCDB	40.4	46.5	57.4	64.7	71.3	78.5	80.4	79.9	71.0	61.7	52.1	41.0	(32)
	10%	WB	35.5	39.0	46.0	52.2	59.2	65.1	67.0	66.1	60.9	54.5	46.7	37.0	(33)
		MCDB	37.4	42.9	53.2	61.8	67.6	75.4	77.4	76.3	67.1	58.5	49.1	38.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	15.6	19.8	20.9	19.9	20.4	20.7	20.1	17.9	15.8	10.5	9.7	(35)
		MCDBR	13.2	22.0	29.3	30.0	28.4	27.3	27.0	27.4	23.5	20.1	14.4	11.8	(36)
		MCWBR	9.0	13.6	16.3	14.5	12.7	10.4	10.0	10.7	10.9	11.5	9.2	8.4	(37)
	5% WB	MCDBR	12.1	20.2	27.5	26.3	25.0	24.8	24.7	24.4	20.9	19.2	13.1	10.8	(38)
		MCWBR	8.6	13.0	15.8	12.9	11.7	9.9	9.8	9.9	9.8	11.4	8.7	8.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.297	0.325	0.368	0.413	0.418	0.424	0.427	0.425	0.394	0.362	0.330	0.297	(40)
	taud		2.394	2.329	2.271	2.207	2.246	2.268	2.261	2.283	2.340	2.402	2.443	2.446	(41)
	Ebn at Noon		249	264	268	266	269	267	264	259	257	247	234	236	(42)
	Edh at Noon		22	29	36	42	42	42	41	39	33	27	21	19	(43)
(44) All-Sky Solar Radiation	RadAvg		381	655	984	1307	1516	1658	1655	1479	1072	737	413	316	(44)
	RadStd		40	81	109	137	171	128	150	171	136	69	59	45	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (109 neighbors)	+0.95	N/A	N/A	+1.57	+0.85	+0.72	-185	-248	+158	+90

Nomenclature: See separate page